C. Powers Of Two

time limit per test 4 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: bitmasks, greedy | https://codeforces.com/problemset/problem/1095/C

A positive integer x is called a *power of two* if it can be represented as $x = 2^y$, where y is a non-negative integer. So, the *powers of two* are 1, 2, 4, 8, 16, ….

You are given two positive integers n and k. Your task is to represent n as the **sum** of **exactly** k *powers of two*.

Input

The only line of the input contains two integers n and k ($1 \le n \le 10^9$, $1 \le k \le 2 \cdot 10^5$).

Output

If it is impossible to represent n as the sum of k powers of two, print `NO`.

Otherwise, print `YES`, and then print k positive integers $b_1, b_2, \ldots, b_k$ such that each of b_i is a power of two, and $\sum_{i=1}^{k} b_i = n$. If there are multiple answers, you may print any of them.

Examples

Input
9 4

Output
YES 1 2 2 4

Input
8 1

Output
YES 8

Input
5 1

Output
NO

Input
3 7

Output
NO